



# REDOCS 2025 - ANSSI subject

## Designing a cryptographic suites scanner for network security protocols

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# What's the problem?

## HTTP communication

## HTTPS (HTTP + TLS) communication

| No. | Time            | Source       | Destination  | Protocol | Length | Info                               |
|-----|-----------------|--------------|--------------|----------|--------|------------------------------------|
| +   | 169.3.184448505 | 10.76.64.86  | 18.236.36.28 | HTTP     | 454    | GET / HTTP/1.1                     |
| +   | 171.3.389325425 | 18.236.36.28 | 10.76.64.86  | HTTP     | 2846   | HTTP/1.1 200 OK (text/html)        |
| +   | 189.3.439240320 | 10.76.64.86  | 18.236.36.28 | HTTP     | 449    | GET /favicon.ico HTTP/1.1          |
| +   | 194.3.693431458 | 18.236.36.28 | 10.76.64.86  | HTTP     | 1622   | HTTP/1.1 404 Not Found (text/html) |

```

Frame 171: 2846 bytes on wire (22768 bits), 2846 bytes captured (22768 bits) on interface wlp2s8f0, id 0
  Ethernet II, Src: Fortinet_09:00:1c:00:09:0f:09:00:1c), Dst: 5c:b4:7e:02:7b:64 (5c:b4:7e:02:7b:64)
    Internet Protocol Version 4, Src: 18.236.36.28, Dst: 10.76.64.86
      Transmission Control Protocol, Src Port: 80, Dst Port: 36384, Seq: 1, Ack: 389, Len: 2780
        Hypertext Transfer Protocol
          Line-based text data: text/html (368 lines)
            \r\n
            <html>\r\n
            <head>\r\n
            \t\t<title>URL for testing.</title>\r\n
            </head>\r\n
            \r\n
            <body>\r\n
            <code>\r\n
            \t\t<a href="http://www.testingmcafeesites.com/index.html">http://www.testingmcafeesites.com/index.html</a><br>\r\n
            \t\t<br>\r\n
            \r\n
            \t\t\tThis is an index url which gives an overview of the different test urls available.<br>\r\n
            \t\t<br>\r\n
            \t\t\t<a href="http://www.testingmcafeesites.com/testcat_ac.html">http://www.testingmcafeesites.com/testcat.ac.</a>\r\n
            \t\t\t\t\tThis is an example URL which should be categorized as an art/culture website with a minimal risk reputa
            \t\t\t\t\t<br>\r\n
            \t\t\t\t\t\r\n
            \t\t\t\t\t<br>\r\n
            \t\t\t\t\t<a href="http://www.testingmcafeesites.com/testcat_al.html">http://www.testingmcafeesites.com/testcat.al.</a>\r\n
            \t\t\t\t\t\t\tThis is an example URL which should be categorized as an alcohol website with a low risk reputation sco
            \t\t\t\t\t\t\t<br>\r\n
            \t\t\t\t\t\t\t\r\n
            \t\t\t\t\t\t\t<br>\r\n
            \t\t\t\t\t\t\t<a href="http://www.testingmcafeesites.com/testcat_an.html">http://www.testingmcafeesites.com/testcat.an.</a>\r\n
            \t\t\t\t\t\t\t\t\tThis is an example URL which should be categorized as an anonymizer website with a low risk reputation
            \t\t\t\t\t\t\t\t\t<br>\r\n
            \t\t\t\t\t\t\t\t\t\r\n

```

```

2144 14.017998073 Fortinet:09:00:1c Broadcast ARP 56 who has 10.76.64.73? Tell 10.76.67.254
2145 14.038982120 185.70.42.37 10.76.64.86 TCP 2522 443 - 42266 [PSH, ACK] Seq=1705 Ack=288
2146 14.031023784 10.76.64.86 185.70.42.37 TCP 66 42266 - 443 [ACK] Seq=288 Ack=4161 Win=1
2147 14.038982189 185.70.42.37 10.76.64.86 TLSv1.2 1006 Application Data
Flags: 0x018 (PSH, ACK)
Window: 83
[Calculated window size: 83]
[Window size scaling factor: -1 (unknown)]
Checksum: 0x37cc [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
[Timestamps]
[SEQ/ACK analysis]
TCP payload (2456 bytes)
[Reassembled PDU in frame: 2147]
TCP segment data (2456 bytes)
0000 5c b4 7e 02 7b 64 00 09 0f 09 00 1c 08 00 45 00 \x5c\xd47e027b6400090f09001c08004500
0001 09 cc 96 0d 40 00 2e 06 7f 11 b9 64 2a 25 04 a8 @.....F%L
0002 40 56 01 bb a5 1a fd 8c 8e fe 3a 8a f7 a2 80 18 @.....[.....
0003 00 53 01 cc 00 00 01 01 88 0a 93 79 d9 98 20 17 ..S.....y
0004 09 b5 17 03 03 0d 01 84 06 29 06 22 9d 4d a8 1 ..9m1k"
0005 27 53 39 3f 7f 79 74 0f 5a 85 49 28 b9 be 06 6 ..S7yqtoZi(8
0006 88 c4 d1 71 ad 06 48 f9 93 14 7f 8c 4e 88 56 7 ..yqtoH.....N\
0007 0b 03 7b 40 41 dd fa df 67 b4 6d 25 a4 44 55 90 ..(A.....dmisD
0008 01 e1 29 71 fd 05 de 13 e2 c3 15 ef 4f 8a 3f d3 ..e1.....?
0009 5a 87 9f f1 f6 f3 c4 c4 f8 cf 08 06 3f 19 26 39 ..Z.....H?Z
000a 4f 46 7a 5f d4 bd 01 e5 f3 ee 08 45 79 0b 5a 39 ..F.....Fy89
000b 22 1f 4d f6 3c 9a 22 7f 9a 2e fa 79 21 ff 64 ..".....y
000c 0b 7d 7c 4a ce 7f 34 59 59 03 df 61 7c 13 08 e7 ..[D.....Y..a]
000d 96 3b cd 19 4b bf 23 ed e4 2a 66 ca 1a 07 ba f8 ..:K.....H?x
000e c7 34 b5 25 bf 67 5a c2 18 01 0b ea 4b 2c 9a 26 ..4%gZ.....&
000f 33 0e c3 49 37 c3 7f 49 44 2a 66 ca 1a 07 ba f8 ..3-I7-I'Df.....d
0010 67 01 a8 b4 e0 37 98 b0 e7 d4 4a 08 60 b3 96 g.....JH
0011 07 b7 ae 27 f1 3c 01 4f 1b f5 60 6d 7c bd 2d 70 ..1.....<O...e-
0012 af 6d 40 8f ce c0 39 a1 b1 21 07 8f ad e1 5b ..no.....9.....
0013 ef db 37 67 47 d3 dd f4 e0 bc e0 f2 1b 81 34 b4 ..7Gg.....K4
0014 13 18 5b 02 0e ac 9a 31 d2 8c 6d 27 0b 4d 88 ..[.....m'kK

```



# Network security protocols

- Confidentiality
- Integrity
- Authentication
- Non-repudiation
- ...



**CRYPTOGRAPHY**



# Network security protocols

- Confidentiality
- Integrity
- Authentication
- Non-repudiation
- ...

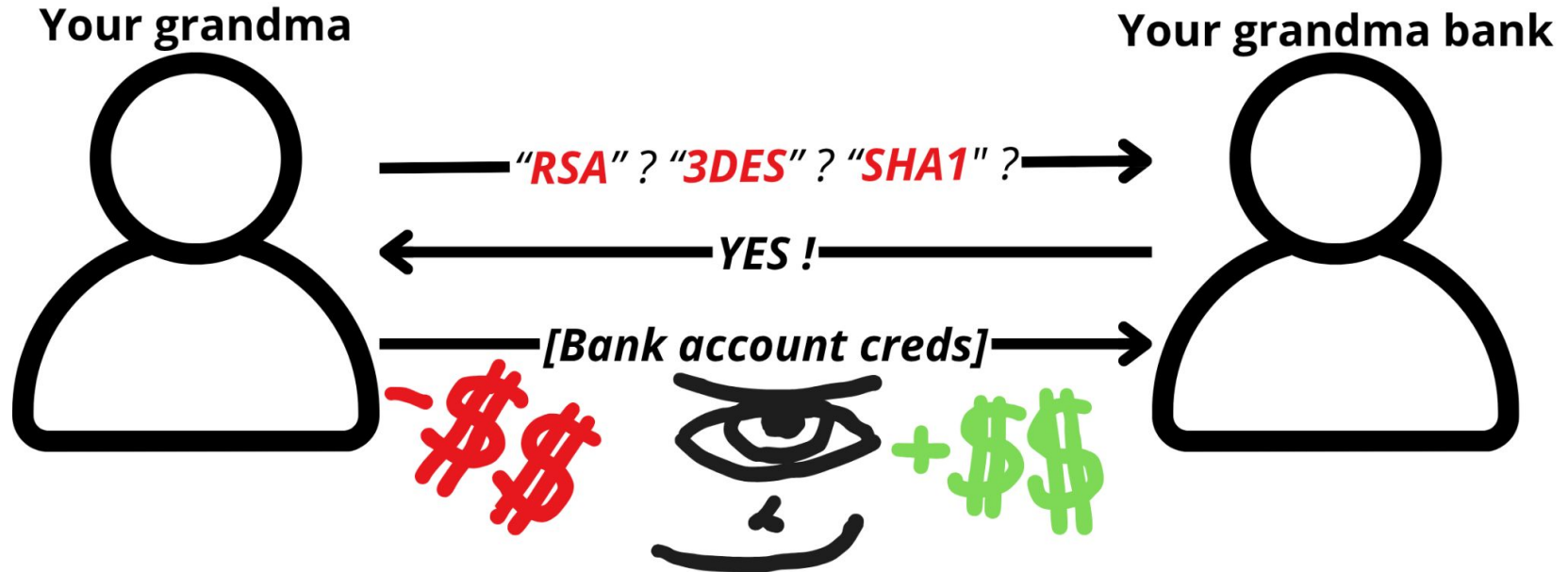
(OLD ???)

CRYPTOGRAPHY

```
CipherSuite TLS_RSA_WITH_RC4_128_SHA  
CipherSuite TLS_RSA_WITH_3DES_EDE_CBC_SHA  
CipherSuite TLS_RSA_WITH_AES_128_CBC_SHA  
= { 0x00, 0x05 };  
= { 0x00, 0x0A };  
= { 0x00, 0x2F };
```

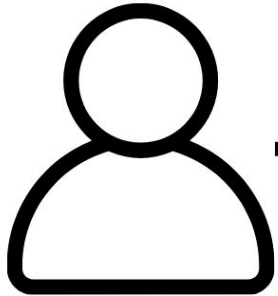
RFC 5246 "TLS v1.2", Aug. 2008

## Why bother?

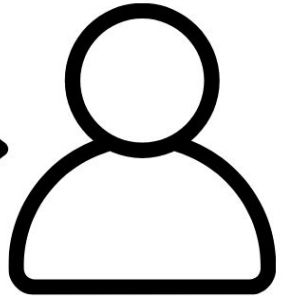




## Why bother? - in-the-middle attack



Can: STRONG\_SUIT,  
WEAK\_SUIT



Can: STRONG\_SUIT,  
WEAK\_SUIT

## Why bother? - in-the-middle attack







# I wouldn't be so dumb

Maybe, maybe not... actually, many targets are vulnerable to an in-the-middle attacks:

| Protocol                           | Vulnerable to Logjam |
|------------------------------------|----------------------|
| HTTPS — Top 1 Million Domains      | 8.4%                 |
| HTTPS — Browser Trusted Sites      | 3.4%                 |
| SMTP+StartTLS — IPv4 Address Space | 14.8%                |
| POP3S — IPv4 Address Space         | 8.9%                 |
| IMAPS — IPv4 Address Space         | 8.4%                 |

*David Adrian, Karthikeyan Bhargavan, Zakir Durumeric, Pierrick Gaudry, Matthew Green, et al.. Imperfect forward secrecy: How Diffie-Hellman fails in practice. Communications of the ACM, 2018, 62 (1), pp.106-114*

# About the ANSSI *(Agence Nationale de la Sécurité des Systèmes d'Information)*

- *Publishes guides and recommendations*
- *Certifies security products*
- *among many other things...*



Produit certifié CC

Publié le 15 Octobre 2025 , mis à jour le 15 Octobre 2025

IDEMIA CA (Version 1.3.1)



## Are there so much protocols?

- TLS
  - v1.1 and older
  - v1.2, v1.3
- SSH
- IPSec
  - IKE
  - EAP
- 802.1X
- Kerberos
- SFTP
- DNSSEC
- VPNs
  - WireGuard
  - IPSec
  - ...
- SNMP
- 5G, 4G, etc.
- Bluetooth
- 3GPP
- *and many other*



## How can one check?

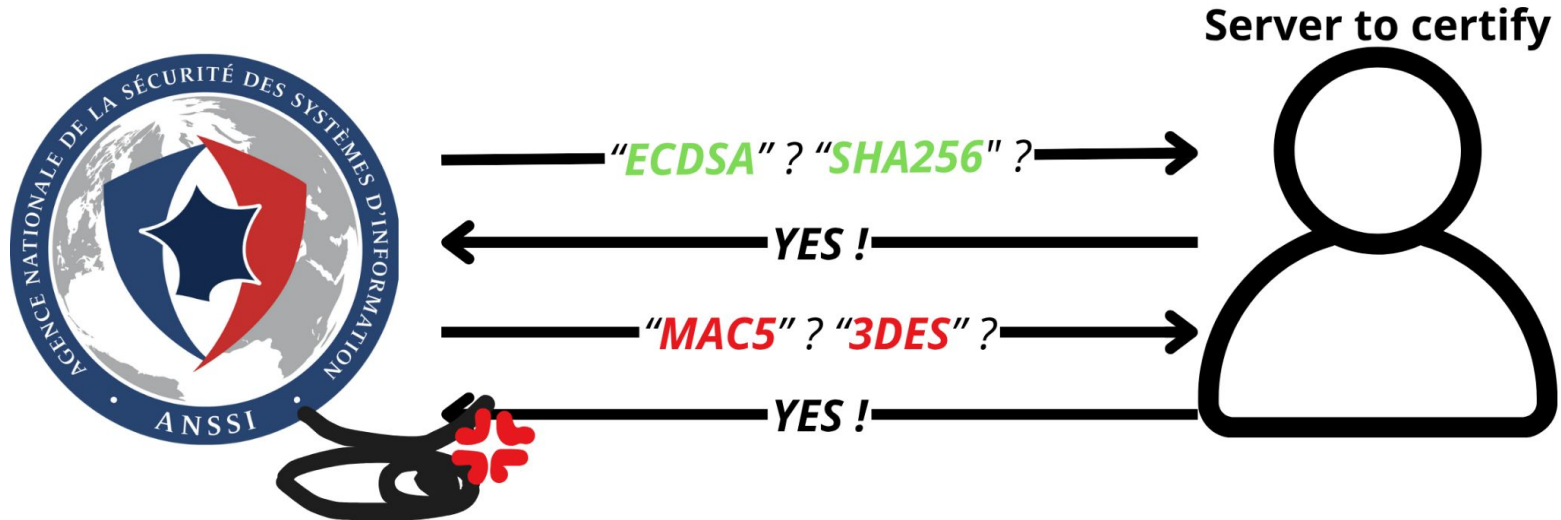
OPTION 1 (White box): Read the source code

- *what are the hardcoded suites, etc.*

OPTION 2 (Black box): You don't have access to the source code.....

- ...let's use ping ?

## Black-box approach



*Every protocol works differently, and implements different cryptographic elements*



## More on that

### Reasons for allowing old crypto

- Retrocompatibility

### Other uses for a multi-protocol scanner

- Observatoire de l'Internet

- Security Operations Center (SOC)

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# Our framework

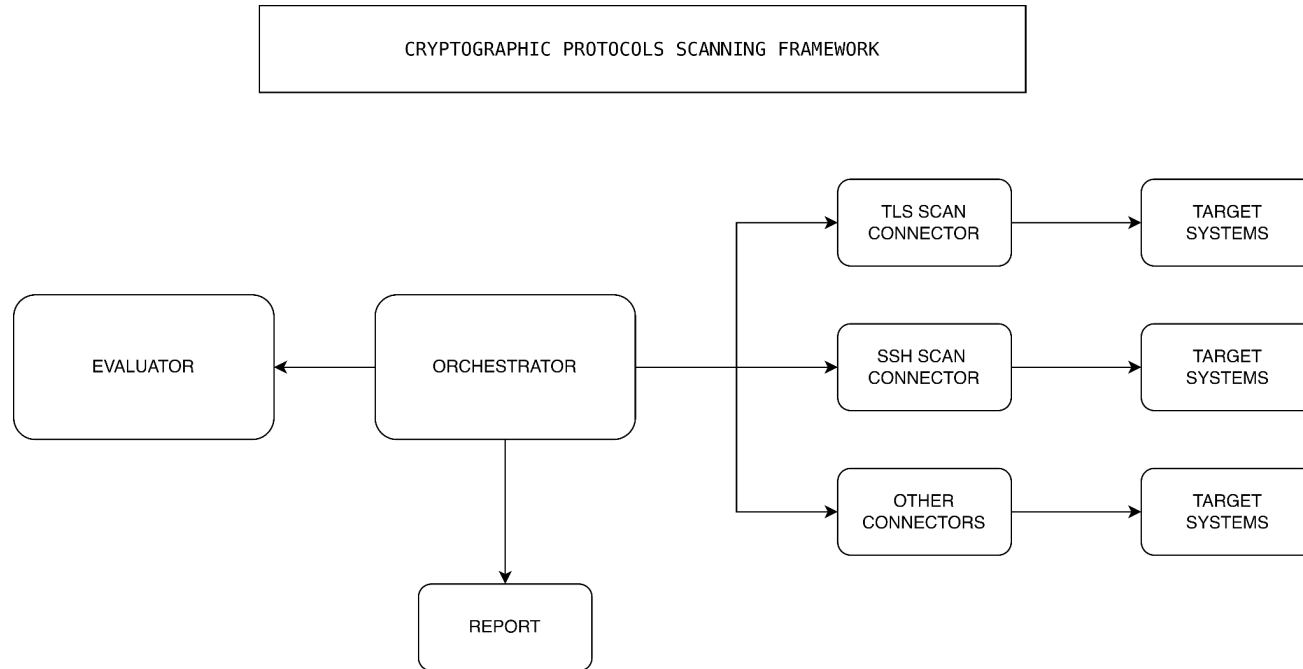


# Key Design Considerations for the Framework

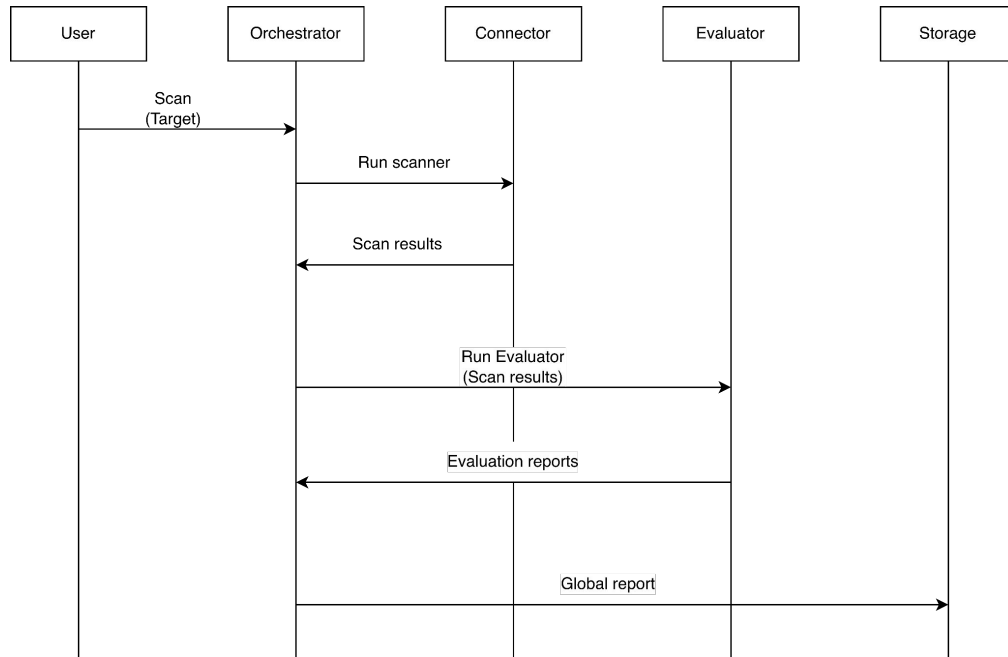
- Modular Architecture - Easy to add new scanners
- Scalable Design - Parallel execution engine
- Extensible Evaluation - Rule-based scoring system
- Enterprise Ready - Security controls and compliance
- Flexible Deployment - Multiple interface options



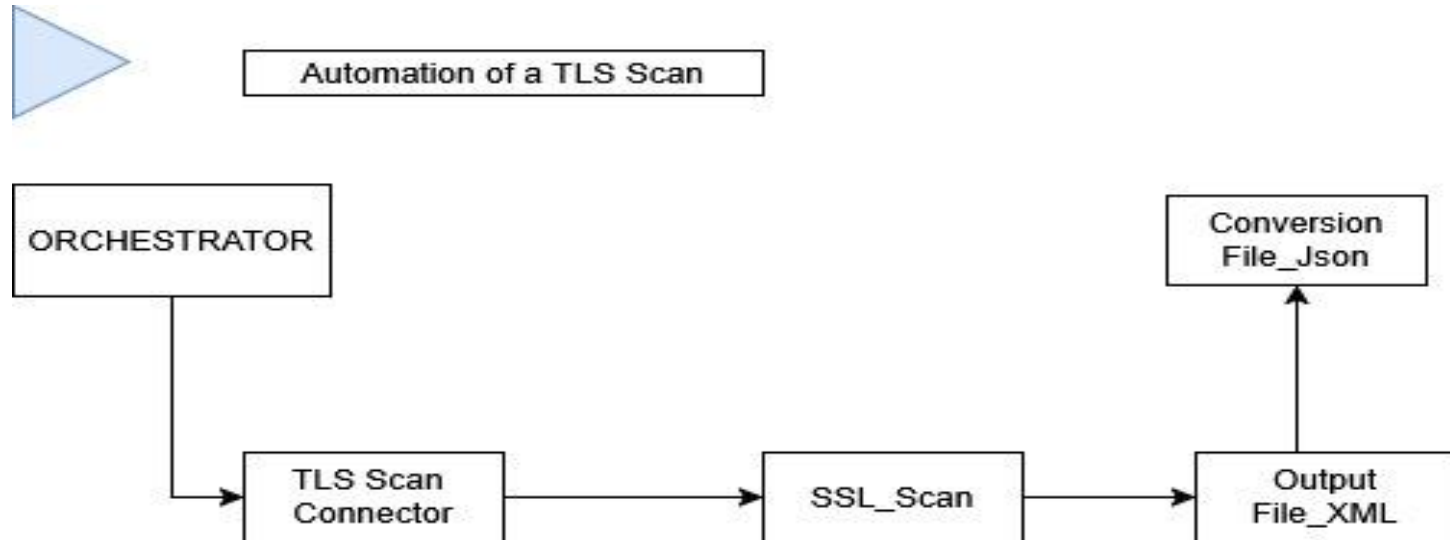
# Architecture diagram



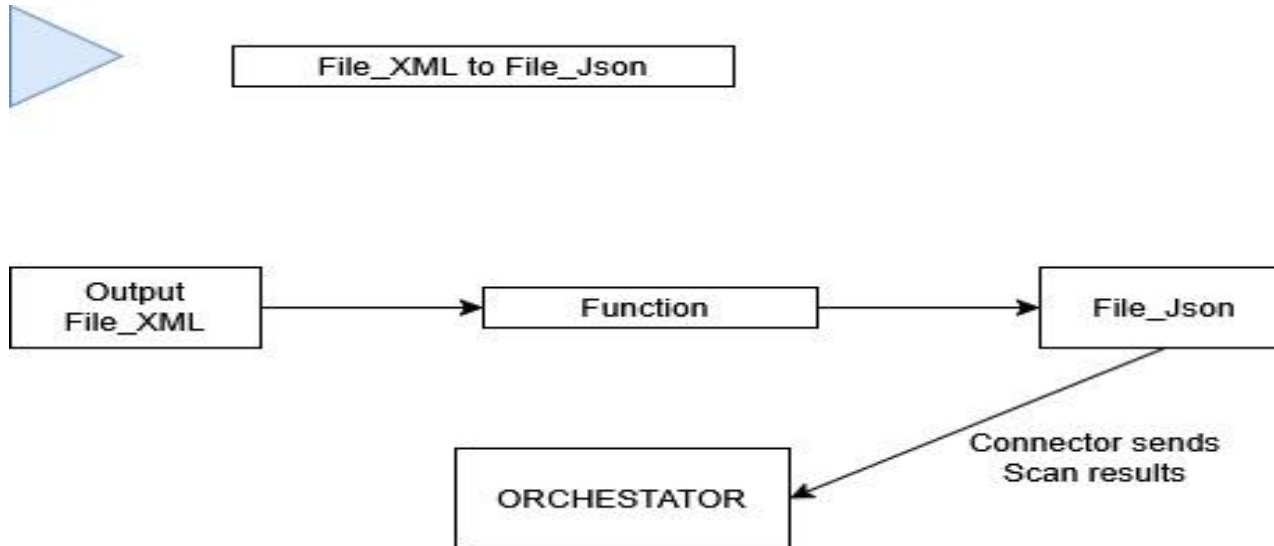
# Framework flow diagram



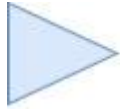
## Orchestrator Coordinates the scan, saves the results



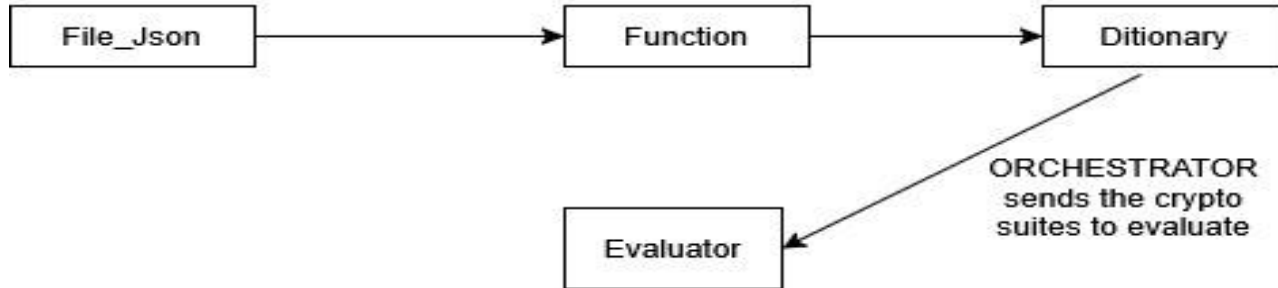
## Orchestrator Coordinates the scan, saves the results



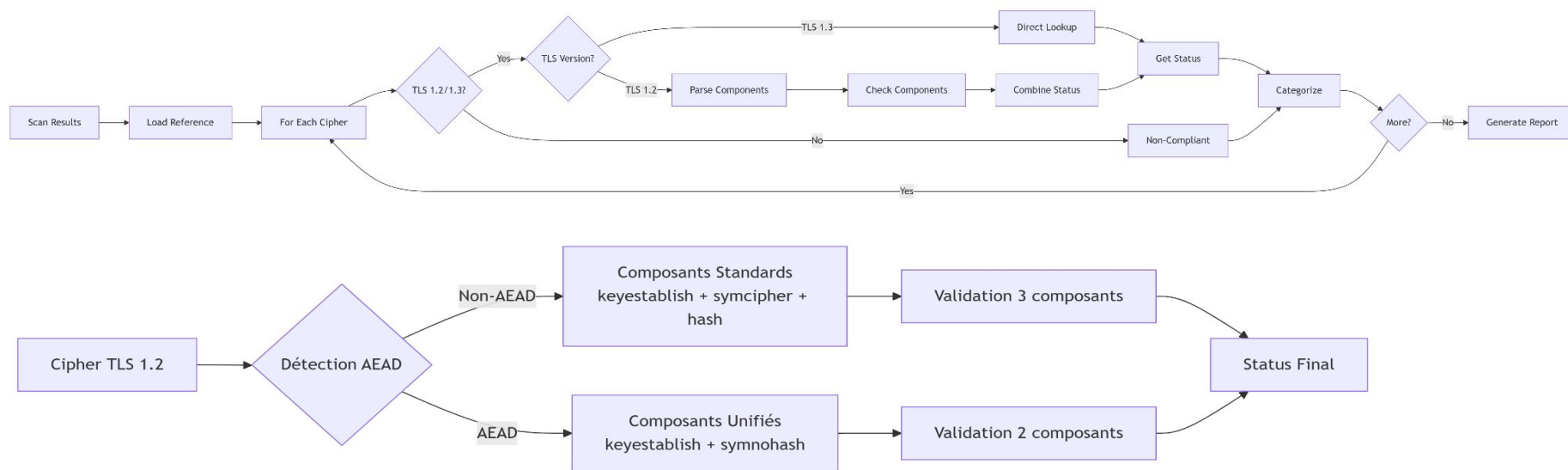
## Orchestrator sends Evaluator



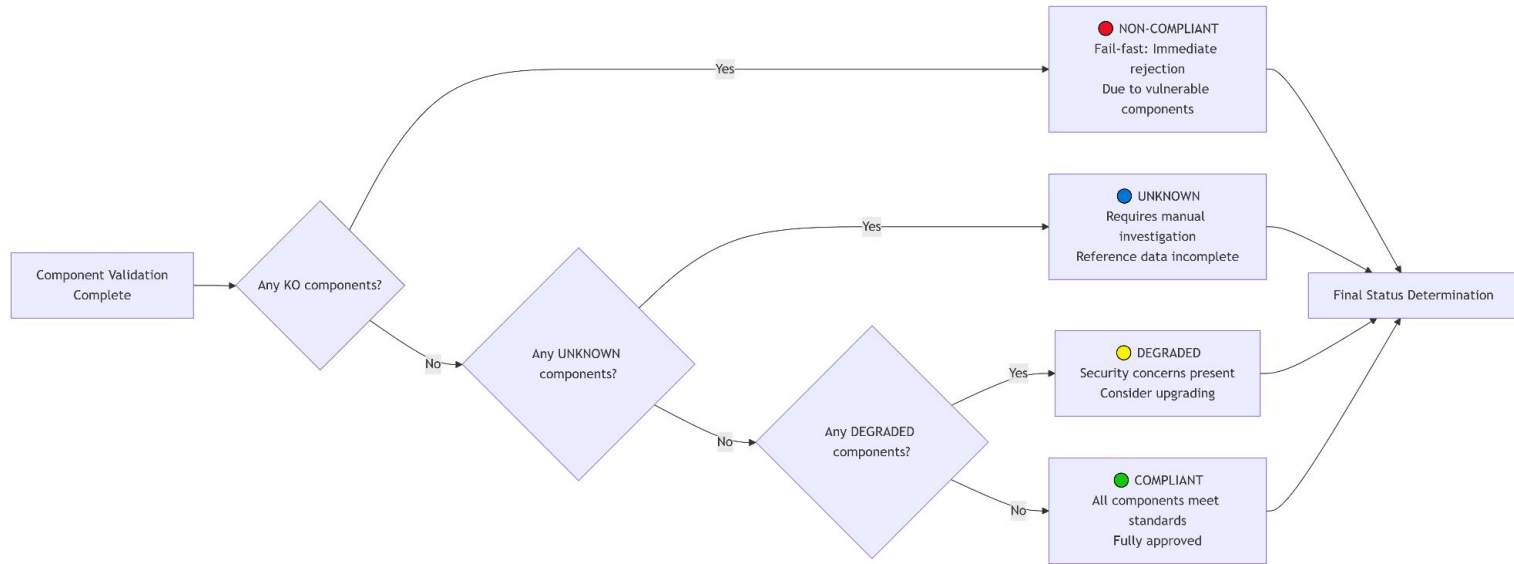
File\_Json to Ditionary



## Compliance Checker Block



## Compliance Checker Block



## Compliance Checker Block

```
71      "TLS_ECDHE_RSA_ARIA_256_GCM_SHA384": {
72        "components": {
73          "ECDHE_RSA": {
74            "status": "OK",
75            "comment": ""
76          },
77          "ARIA_256_GCM": {
78            "status": "DEGRADED",
79            "comment": "The use of AES is advertised"
80          },
81          "SHA384": {
82            "status": "OK",
83            "comment": ""
84          }
85        },
86        "final_status": "DEGRADED",
87        "comment": "One or more components degraded: ARIA_256_GCM"
88      },
89      "TLS_ECDHE_RSA_AES_128_GCM_SHA256": {
90        "components": {
91          "ECDHE_RSA": {
92            "status": "OK",
93            "comment": ""
94          },
95          "AES_128_GCM": {
96            "status": "OK",
97            "comment": ""
98          },
99          "SHA256": {
100            "status": "OK",
101            "comment": ""
102          }
103        },
104        "final_status": "COMPLIANT",
105        "comment": "All components compliant"
106      },
```

```
1      "TLSv1.3": {
2        "TLS_AES_128_GCM_SHA256": {
3          "components": {
4            "TLS_AES_128_GCM_SHA256": {
5              "status": "OK",
6              "comment": ""
7            }
8          },
9          "final_status": "COMPLIANT",
10         "comment": ""
11       },
12       "TLS_AES_256_GCM_SHA384": {
13         "components": {
14           "TLS_AES_256_GCM_SHA384": {
15             "status": "OK",
16             "comment": ""
17           }
18         },
19         "final_status": "COMPLIANT",
20         "comment": ""
21       },
22       "TLS_CHACHA20_POLY1305_SHA256": {
23         "components": {
24           "TLS_CHACHA20_POLY1305_SHA256": {
25             "status": "OK",
26             "comment": ""
27           }
28         },
29         "final_status": "COMPLIANT",
30         "comment": ""
31       },
32     },
33   },
```



---

# Transport Layer Security (TLS)



# TLS – Objectives

## OBJECTIVES :

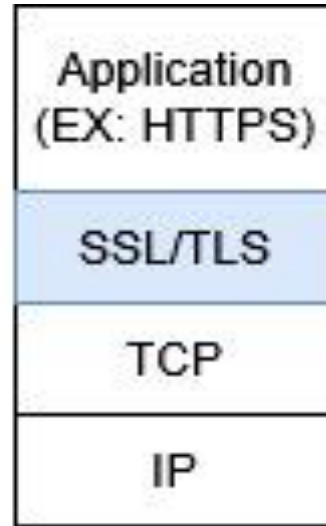
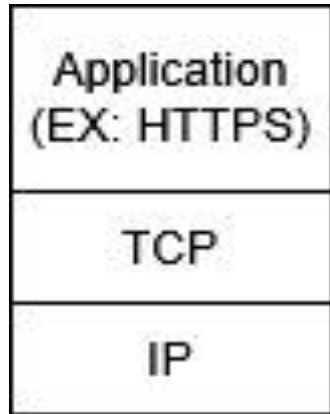
- 1) **Confidentiality** between two applications: symmetric cryptography (AES, ...)
- 2) **Integrity of communication**: message MAC (Message Authentication Code)

## Functions :

- 1) **Peer authentication**: use of asymmetric cryptography for optionally mutual authentication
- 2) **Negotiation of the shared secret** : to ensure the security of the communication
- 3) **Secure negotiation** : Any modification in the exchange must be detected

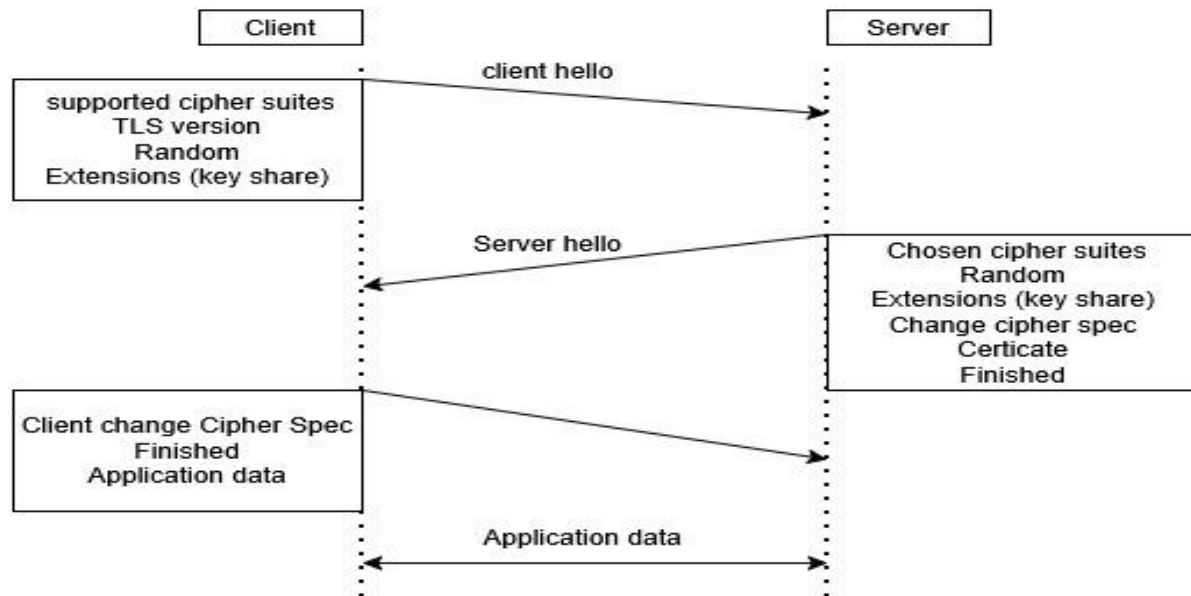


## Position of SSL/TLS

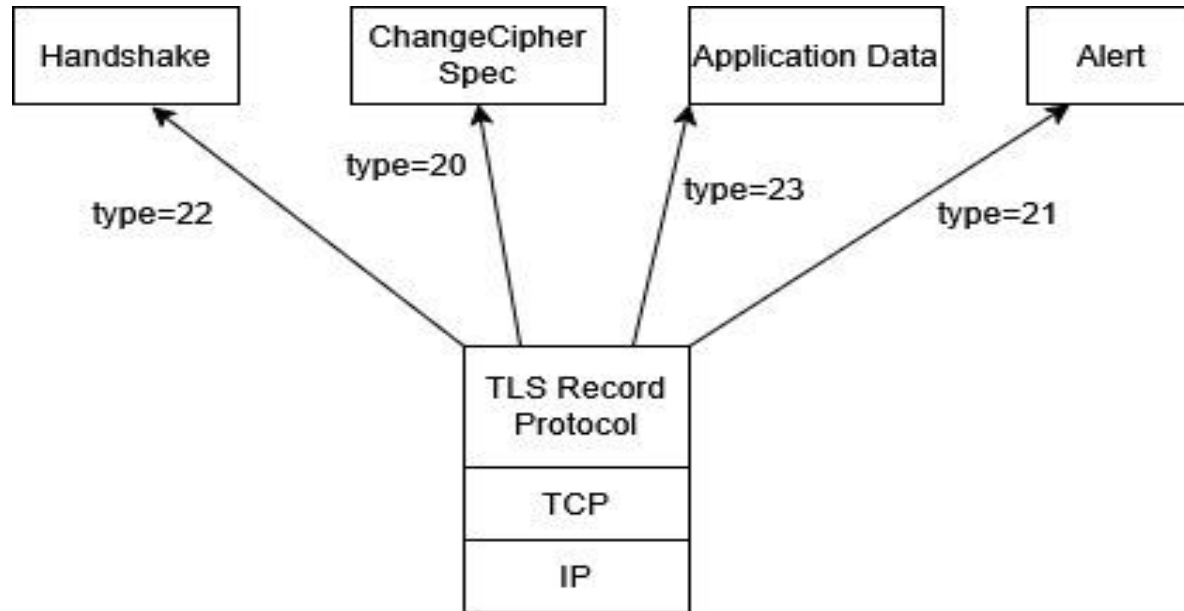


# TLS Connection

## Handshake



# TLS Encapsulation





## (Reminder) TLS Cipher suites

Example of a cipher suite:

TLS\_ECDHE\_ECDSA\_WITH\_AES\_256\_CBC\_SHA384

|                          |             |
|--------------------------|-------------|
| → Key Exchange Mechanism | ECDHE-ECDSA |
| → Symmetric Cypher       | AES-256-CBC |
| → Hash                   | SHA-384     |



## Evaluation by ANSSI standard

### Criteria ?

- Algorithm itself
- Parameters:
  - Modes, curves, key size
- Known attacks

### 3 Levels of security:

- Recommended
- Degraded
- K.O.

See: PG-083 and PA-079 standards



## Our reference file

Recover all components used in TLS ciphersuites, see [iana.com](https://iana.com)

All TLS...:

- Key exchange mechanism
- Symmetric cipher
- Hash function
- Group
- Connection signature

```
"3DES_EDE_CBC": {  
  "status": "KO",  
  "comment": "Insecure: Vulnerabilities"  
},  
"AES_128_CBC": {  
  "status": "DEGRADED",  
  "comment": "Insecure: Padding oracle"  
},
```



---

# Secure SHell (SSH)



# SSH Protocol

- SSH is an applicative layer security suite that secures remote shell connections — it provides:
  - Confidentiality,
  - Integrity
  - Double authentication:
    - Signing messages
    - Logging in the system
- Allows for tunneling and port forwarding
- By default: ranking of all allowed cipher suites is sent by the server





## Existing tool into our framework

An open source repository that gives an all-in-one script:

→ Reshape functions into fitting into the framework

- ◆ Connector
- ◆ Evaluator
- ◆ Reference file

→ Components of SSH cipher suites:

- ◆ Key Exchange
- ◆ Symmetric Cipher
- ◆ Macs
- ◆ Host Key Algorithm

---

# Internet Protocol Security (IPSec)



# What is IPsec?

- IPsec is a network layer security suite that secures IP packets — it provides:
  - Confidentiality (encryption),
  - Integrity (tamper detection),
  - Authentication (who's who),
  - Anti-replay protection (no reuse of captured packets).
- It runs at Layer 3 (network layer) of the OSI model — below TCP/UDP.



## IPsec Architecture: main parts

| Plane         | Protocols                            | Purpose                                                |
|---------------|--------------------------------------|--------------------------------------------------------|
| Control Plane | IKEv1 / IKEv2                        | Negotiates keys, algorithms, authentication, policies. |
| Data Plane    | ESP (Encapsulating Security Payload) | Encrypts and authenticates actual IP packets.          |

- **SA (security associates)** “set of parameters and keys”.
  - **IKE SA** → defines how the control plane messages are protected.
  - **CHILD SA** → defines how the ESP packets are protected.



# IKE (Internet Key Exchange)

- IKE = key management protocol,
- Runs over UDP port 500
- Establishes secure tunnels by negotiating:
  - Encryption algorithms (AES, 3DES...)
  - Integrity algorithms (HMAC-SHA2, etc.)
  - Diffie-Hellman groups (for key exchange)
  - Authentication method (certificates, PSK...)
  - Lifetime, traffic selectors, etc.
- IKE works in two phases (in IKEv1) or as two exchanges (in IKEv2):

| Phase                 | What it creates | What it's for                          |
|-----------------------|-----------------|----------------------------------------|
| IKE SA<br>(Phase 1)   | Control channel | Secure tunnel for negotiation messages |
| CHILD SA<br>(Phase 2) | Data channel    | Protects actual IP packets (ESP/AH)    |





# IKE-Scan

| Sent by ike-scan:                                                                                                                                                         | Response from Server:                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Initiator Cookie: 0x123456789ABCDEF0<br>Responder Cookie: 0x0000000000000000<br>Exchange Type: Main Mode (2)<br>Payload: SA (proposals: Encryption, Hash, Auth, DH Group) | Initiator Cookie: 0x123456789ABCDEF0<br>Responder Cookie: 0xFEDCBA9876543210<br>Exchange Type: Main Mode (2)<br>Payload: SA (selected proposal: Encryption, Hash, Auth, DH Group)<br>Vendor ID: Cisco-Unity |

```
root@aa46a12f5247:~# ike-scan -2 192.168.0.2
Starting ike-scan 1.9.6 with 1 hosts (http://www.nta-monitor.com/tools/ike-scan/)
192.168.0.2    IKEv2 SA_INIT Handshake returned HDR=(CKY-R=e288aa363d673646, IKEv2) SA=(Encr=3DES Integ=HMAC_MD5_96 Prf=H
MAC_MD5 DH_Group=2:modp1024) KeyExchange(132 bytes) Nonce(32 bytes) CertificateRequest(21 bytes) Notification(4 bytes) Not
ification(4 bytes)
```



## Problem with IKE-Scan

- Single Proposal: ike-scan 192.168.1.1
- Multiple Proposals: ike-scan --trans=1,3,2,1,1 --trans=1,3,2,2,1 192.168.1.1

```
Starting ike-scan 1.9.6 with 1 hosts (http://www.nta-monitor.com/tools/ike-scan/)
192.168.0.2    IKEv2 SA_INIT Handshake returned HDR=(CKY-R=e288aa363d673646, IKEv2) SA=(Encr=3DES Integ=HMAC_MD5_96 Prf=H
MAC_MD5 DH_Group=2:modp1024) KeyExchange(132 bytes) Nonce(32 bytes) CertificateRequest(21 bytes) Notification(4 bytes) Not
ification(4 bytes)
```



# IKEv2 Suite Scanner

- Purpose:
  - Test all IKEv2 crypto suite **combinations** against a VPN server.
  - Identify supported/unsupported suites and assess security.
- Features:
  - Automates ike-scan over all combinations of Encryption, PRF, Integrity, DH.
  - Logs reason for rejection (e.g., weak cipher, unsupported DH).
  - Marks each primitive:  Secure /  Caution /  Weak.
  - Outputs accepted suites with readable crypto names.

# Crypto Algo Combinations

```
const id_name_map encr_map[] = {
    {1, "DES_IV64"},
    {2, "DES"},
    {3, "3DES"},
    {4, "RC5"},
    {5, "IDEA"},
    {6, "CAST"},
    {7, "Blowfish"},
    {8, "3IDEA"},
    {9, "DES_IV32"},
    {11, "NULL"},
    {12, "AES_CBC"},
    {13, "AES_CTR"},
    {14, "AES_CCM_ICV8"},
    {15, "AES_CCM_ICV12"},
    {16, "AES_CCM_ICV16"},
    {18, "AES_GCM_ICV8"},
    {19, "AES_GCM_ICV12"},
    {20, "AES_GCM_ICV16"},
    {23, "CAMELLIA_CBC"},
    {-1, NULL}
```

```
const id_name_map prf_map[] = {
    {1, "HMAC_MD5"},
    {2, "HMAC_SHA1"},
    {3, "HMAC_TIGER"},
    {4, "AES128_XCBC"},
    {5, "HMAC_SHA2_256"},
    {6, "HMAC_SHA2_384"},
    {7, "HMAC_SHA2_512"},
    {8, "HMAC_AES128_CMAC"},
    {-1, NULL}
```

```
const id_name_map auth_map[] = {
    {1, "PSK"},
    {2, "DSS"},
    {3, "RSA_Sig"},
    {4, "RSA_Enc"},
    {5, "RSA_RevEnc"},
    {6, "ElGamel_Enc"},
    {7, "ElGamel_RevEnc"},
    {8, "ECDSA_Sig"},
    {9, "ECDSA_SHA256"},
    {10, "ECDSA_SHA384"},
    {11, "ECDSA_SHA512"},
    {128, "CRACK"},
    {64221, "Hybrid_RSA"},
    {64223, "Hybrid_DSS"},
    {65001, "XAUTH_PSK"},
    {65003, "XAUTH_DSS"},
    {65005, "XAUTH_RSA"},
    {65007, "XAUTH_RSA_Enc"},
    {65009, "XAUTH_RSA_RevEnc"},
    {-1, NULL}
```

## Output Multiple Suites

```
20      | 7  | 14      | 19      | ❌ Not Supported | Responder did not accept this combination. | AES-GCM_ICV16, HM
AC-SHA2-512, HMAC-SHA2-512-256, ecp256 AES-GCM: ✅ Strong. PRF=HMAC-SHA2: ✅ Strong. Integrity=HMAC-SHA2: ✅ Strong. DH=EC
P: ✅ Modern.
Testing: Encr=20, PRF=7, Integ=14, DH=20
20      | 7  | 14      | 20      | ❌ Not Supported | Responder did not accept this combination. | AES-GCM_ICV16, HM
AC-SHA2-512, HMAC-SHA2-512-256, ecp384 AES-GCM: ✅ Strong. PRF=HMAC-SHA2: ✅ Strong. Integrity=HMAC-SHA2: ✅ Strong. DH=EC
P: ✅ Modern.
Testing: Encr=23, PRF=2, Integ=2, DH=2
23      | 2  | 2        | 2        | ❌ Not Supported | Responder did not accept this combination. | CAMELLIA_CBC, HMA
C-SHA1, HMAC-SHA1-96, modp1024 CAMELLIA: ⚠️ Secure but less common. PRF=HMAC-SHA1: ⚠️ Deprecated. Integrity=HMAC-SHA1-96:
⚠️ Weak. DH=modp1024: ❌ Weak.
Testing: Encr=23, PRF=2, Integ=2, DH=14
23      | 2  | 2        | 14      | ❌ Not Supported | Responder did not accept this combination. | CAMELLIA_CBC, HMA
C-SHA1, HMAC-SHA1-96, modp2048 CAMELLIA: ⚠️ Secure but less common. PRF=HMAC-SHA1: ⚠️ Deprecated. Integrity=HMAC-SHA1-96:
⚠️ Weak. DH=modp2048: ⚠️ Acceptable.
```



## Result

```
=====
✅ Supported Suites Summary (Human-readable):
Encryption | PRF | Integrity | DH
AES-CBC | HMAC-SHA1 | HMAC-SHA1-96 | modp1024
AES-CBC | HMAC-SHA1 | HMAC-SHA1-96 | modp2048
AES-CBC | HMAC-SHA1 | HMAC-SHA1-96 | modp3072
AES-GCM_ICV8 | HMAC-SHA1 | HMAC-SHA1-96 | ecp256
AES-GCM_ICV8 | HMAC-SHA1 | HMAC-SHA1-96 | ecp384
AES-GCM_ICV8 | HMAC-SHA1 | HMAC-SHA2-256-128 | modp1024
AES-GCM_ICV16 | HMAC-SHA2-256 | HMAC-SHA2-384-192 | ecp384
AES-GCM_ICV16 | HMAC-SHA2-256 | HMAC-SHA2-512-256 | modp1024
AES-GCM_ICV16 | HMAC-SHA2-256 | HMAC-SHA2-512-256 | modp2048
=====
Total suites checked: 576
Total suites accepted: 9
```

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# Future Work



# What we are planning

Extended Protocol Support

Enhanced Scanning Capabilities

Intelligence & Automation

Platform Expansion

Global Compliance



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# Conclusion



## We win

"In today's digital landscape, cryptographic security is not just a technical requirement, it's a business imperative. Our framework transforms this complex challenge from a periodic audit burden into a continuous, automated, and actionable security advantage."